Pentose phosphate pathway

The **pentose phosphate pathway** (also known as **phosphogluconate pathway** or **hexose monophosphate pathway**) is an alternative pathway to glycolysis for glucose oxidation, an alternative pathway to glycolysis for glucose oxidation, whose main function is not ATP production. The goals of this route are:

1. Generation of NADPH

NADPH acts as a carrier and donor of reducing equivalents and provides reducing power for biosynthetic reactions. **The pentose phosphate pathway is a major cytosolic source of NADPH**.

The pentose phosphate pathway is important in **tissues where fatty acid and steroid biosynthesis is actively taking place**, such as the mammary gland, adipose tissue, adrenal cortex, or liver. Lipid biosynthesis requires NADPH to reduce the more oxidized precursors.

NADPH also provides the reducing power required for the conversion of ribonucleotides into deoxyribonucleotides during DNA synthesis. Therefore, **rapidly proliferating cells**, such as those of the bone marrow, skin, and intestinal mucosa, have high pentose phosphate pathway activity because they require both reducing power and ribose 5-phosphate for nucleotide synthesis.

Finally, NADPH is also essential for antioxidant defense. It provides the reducing power required by glutathione reductase to regenerate reduced glutathione (GSH) from oxidized glutathione (GSSG). GSH is then used to eliminate reactive oxygen species (ROS) such as hydrogen peroxide. This function is particularly important in **erythrocytes**, which are continuously exposed to oxygen and depend on the pentose phosphate pathway as their main source of NADPH.

2. Synthesis of pentoses

The pentose phosphate pathway allows the synthesis of pentoses from glycolytic/gluconeogenic precursors. In turn, pentoses are important precursors for the synthesis of other molecules. For example, ribose 5-phosphate provides the ribose moiety required for the synthesis of nucleotides and nucleotide-derived molecules such as ATP, CoA, NAD^+ and FAD.

3. Recycling of pentoses

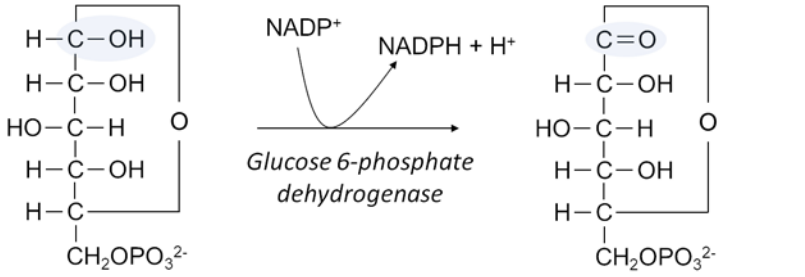
Through the pentose phosphate pathway, pentoses are converted into hexoses and trioses, so that they can be incorporated into the glycolytic or the gluconeogenic pathways.

The pentose phosphate pathway takes place in the cytosol and is divided into two phases:

- **Oxidative phase**: an irreversible process that produces NADPH and ribulose 5-phosphate.
- **Non-oxidative phase**: a reversible series of reactions that interconverts pentose phosphates and glycolytic/gluconeogenic intermediates.

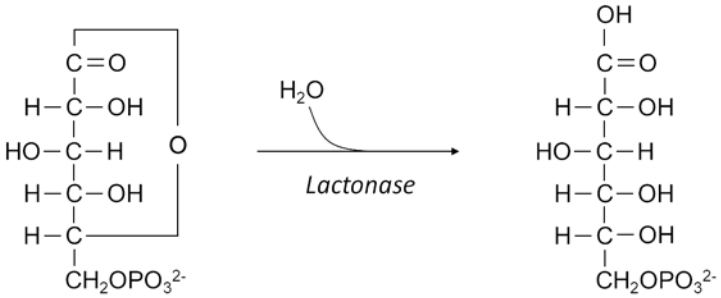
Oxidative phase

The first reaction of the pentose phosphate pathway is the **oxidation of glucose 6-phosphate** in a reaction catalyzed by ***glucose 6-phosphate dehydrogenase***. In this reaction, a molecule of NADPH is formed.



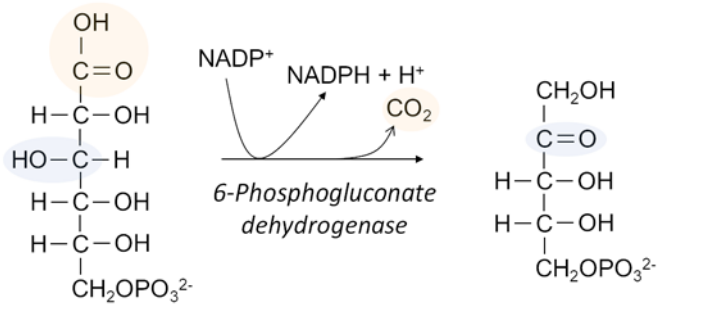
Glucose 6-phosphate — 6-Phosphoglucono-δ-lactone

Next, the **lactone ring is hydrolyzed** to 6-phosphogluconate in a reaction catalyzed by ***lactonase***.



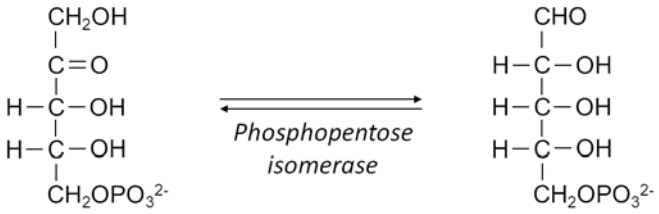
6-Phosphoglucono-δ-lactone — 6-Phosphogluconate

In the third reaction, ***6-phosphogluconate dehydrogenase*** catalyzes the **oxidative decarboxylation of 6-phosphogluconate** to ribulose 5-phosphate. Thus, a five-carbon sugar is produced from a six-carbon sugar, and a second molecule of NADPH is generated.



6-Phosphogluconate — Ribulose 5-phosphate

Ribulose 5-phosphate can be isomerized into ribose 5-phosphate in a reaction catalyzed by ***phosphopentose isomerase***.



Ribulose 5-phosphate Ribose 5-phosphate

If a cell requires both NADPH and ribose 5-phosphate, the products of the oxidative phase can be used directly for biosynthetic purposes. The net balance of these reactions is:

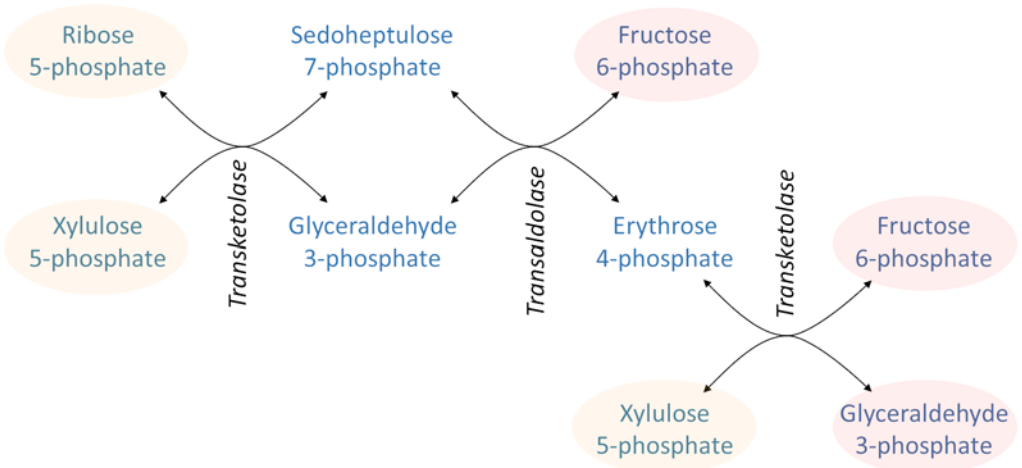
$$\text{Glucose 6-phosphate} + 2\ NADP^+ + H_2O \rightarrow \text{ribose 5-phosphate} + CO_2 + 2\ NADPH + 2\ H^+$$

Non-oxidative phase

The second phase of the pentose phosphate pathway is the non-oxidative synthesis of glycolytic intermediates from pentoses. In particular, **two hexose phosphates and one triose phosphate are formed from three pentose phosphates** (2·6C + 3C = 3·5C). The reactions in this phase are easily reversible, so pentoses can also be synthesized from glycolytic/gluconeogenic intermediates.

The three pentoses that feed the pathway are **one molecule of ribose 5-phosphate and two molecules of xylulose 5-phosphate** (which can be obtained from ribulose 5-phosphate by the action of *ribulose 5-phosphate epimerase*). Through three carbon-transfer reactions, **two molecules of fructose 6-phosphate and one molecule of glyceraldehyde 3-phosphate** are produced.

The three carbon-transfer reactions are catalyzed by two enzymes: ***transketolase***, which uses **thiamine pyrophosphate** (from thiamine or vitamin B_1) as a coenzyme and ***transaldolase***. *Transketolase* transfers two-carbon units and catalyzes the first and third reactions of the process. *Transaldolase* transfers a three-carbon unit and catalyzes the second reaction.



The balance of the non-oxidative phase of the pentose phosphate pathway is:

2 Xylulose 5-phosphate + ribose 5-phosphate

→ 2 fructose 6-phosphate + glyceraldehyde 3-phosphate

The reversibility of the non-oxidative phase gives the pentose phosphate pathway considerable metabolic flexibility. Depending on cellular needs, pentose phosphates can be converted into glycolytic intermediates, or glycolytic intermediates can be used to generate pentose phosphates without producing NADPH.

Global balance of the pentose phosphate pathway

3 Glucose 6-phosphate + 6 $NADP^+$ + 3 H_2O

→ 2 fructose 6-phosphate + glyceraldehyde 3-phosphate + 6 NADPH + 6 H^+ + 3 CO_2

Regulation of the pentose phosphate pathway

Oxidative phase

The oxidative phase of the pentose phosphate pathway is mainly controlled at the level of glucose 6-phosphate dehydrogenase. Its activity is strongly influenced by the **$NADP^+$/NADPH ratio**. $NADP^+$ is a substrate of the enzyme, whereas **NADPH inhibits the activity of glucose 6-phosphate dehydrogenase**. Therefore, when NADPH is consumed, the concentration of NADP**+** increases and glucose 6-phosphate dehydrogenase activity rises, accelerating the oxidative phase of the pathway. This provides an important mechanism for responding to oxidative stress. For example, the reduction of oxidized glutathione (GSSG) to reduced glutathione (GSH) consumes NADPH and generates $NADP^+$, thereby stimulating NADPH production through the pentose phosphate pathway.

Non-oxidative phase

The enzymes of the non-oxidative phase are not subject to major allosteric regulation. Because the reactions are reversible, the direction and rate of the pathway depend mainly on the concentrations of substrates and products.

Deficiencies associated with the pentose phosphate pathway

Thiamine deficiency

Thiamine (vitamin B_1) is the precursor of **thiamine pyrophosphate** (TPP), the cofactor of *transketolase*. TPP is also a cofactor for other important enzymes in our metabolism, including *pyruvate dehydrogenase* and α-*ketoglutarate dehydrogenase*.

Severe thiamine deficiency can cause **beriberi** and **Wernicke–Korsakoff syndrome**. Wernicke encephalopathy is particularly associated with chronic alcohol misuse and is characterized mainly by confusion, ataxia, and abnormalities of eye movement. If untreated, it can progress to Korsakoff syndrome, characterized by severe memory impairment and confabulation. Chronic alcohol misuse is an important risk factor because it is frequently associated with poor thiamine intake and impaired thiamine absorption and metabolism.

Genetic deficiencies affecting the pentose phosphate pathway

The most frequent genetic alteration of the pentose phosphate pathway is the **deficiency in the enzyme *glucose 6-phosphate dehydrogenase*** (G6PD). This defect results in a decrease in the production of NADPH. It is one of the most frequent enzyme deficiencies known (400 million people worldwide are affected, with frequencies of up to 25% in parts of Africa and the Middle East). G6PD deficiency is an X-linked genetic disorder and is therefore more frequently expressed in males. The deficiency reduces the enzyme activity. The degree of reduction varies between genetic variants, but all of them maintain some activity (a complete elimination of G6PD activity would be lethal).

Most individuals with G6PD deficiency are asymptomatic under normal conditions. However, when cellular oxidative stress increases (e.g., due to infections, certain drugs, or the ingestion of fava beans), G6PD-deficient cells are not able to produce enough NADPH to maintain redox homeostasis.

Erythrocytes are particularly vulnerable to G6PD deficiency. Because they lack mitochondria, the pentose phosphate pathway is their only source of NADPH. NADPH is required to maintain glutathione in its reduced form and thereby protect hemoglobin and other cellular components from oxidative damage. Therefore, exposure to strong oxidative stress can cause acute hemolytic anemia in individuals with G6PD deficiency.

Author: Ruben Pio

This work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License. To view a copy of this license, visit http://creativecommons.org/licenses/by-nc-sa/4.0/.